

Problem Set 4: How much do receptors compete?

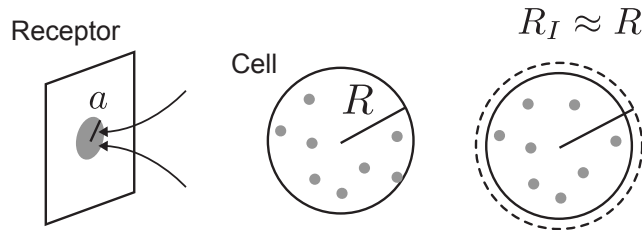
Physics 230A/146A — Biological Physics

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A sphere of radius R that is a perfect absorber, in a far-field concentration c_∞ of particles with diffusion coefficient D , will absorb those particles at a rate $\lambda_{\text{perfect}} = 4\pi DRc_\infty$. This implies that a cell of radius R that were totally covered in receptors for a particular ligand would interact with the ligand at rate λ_{perfect} .

In reality, only a small fraction of the surface of a cell is covered by receptors for a particular ligand. In this problem, we compute the interaction rate for a cell with partial absorption.

Assume the cell has radius $R \sim 10\ \mu\text{m}$ and is covered by N receptors, each of which is a disk of radius $a \sim 10\ \text{nm}$. We make use of an approximation method that assumes there is a radius $r = R_I$ that is sufficiently far from the cell surface, compared to the size of a receptor, that it appears infinitely far at the scale of this receptor, but is sufficiently close to the cell surface that, from far away, it appears approximately the same size as the cell. (By analogy, the planet Earth is of radius R , and people on its surface are of size $a \ll R$. Let R_I be the radius of the Earth's stratosphere: At the scale of a person, R_I is very far from the person, yet at the scale of the solar system, the radius of the Earth with or without the stratosphere are indistinguishable, i.e., $R_I \approx R$.)



- i. An infinite sheet that is no-flux, except for a perfect absorbing disk of radius a , absorbs particles at rate $\lambda_{\text{disk}} = 4Dac_I$ where c_I is the concentration sufficiently far from the disk. Therefore the rate at which N disks absorb particles is $\lambda_N = 4DNac_I$.

At $r = R_I$, what is the average flux (in particles per time per unit area) of particles, $\langle J(R_I) \rangle$? The answer should involve c_I (which is unspecified at this stage in the calculation).

- ii. At the scale larger than the cell, the concentration $c(r)$ satisfies the steady-state diffusion equation

$$0 = D\nabla^2 c.$$

This equation has 3 boundary conditions:

$$\begin{aligned} c(r) &\rightarrow c_\infty & \text{as } r &\rightarrow \infty \\ c(R_I) &= c_I, \\ J(R_I) &= -D\nabla c. \end{aligned}$$

Note that this type of ODE usually requires 2 boundary conditions, but since c_I is unspecified, here 3 boundary conditions are needed. Solve the system by making two approximations: First, assume $J(R_I) = \langle J(R_I) \rangle$ from part (i). (This corresponds to assuming the stratosphere is far from the receptors.) Second, assume $R = R_I$ (This corresponds to assuming the stratosphere is close to the surface).

- iii. What is the rate λ at which particles arrive at the surface? This will depend on N . You should find that as $N \rightarrow \infty$, the rate $\lambda \rightarrow \lambda_{\text{perfect}}$, the rate for a perfectly absorbing sphere.
- iv. How many receptors N are needed to obtain half the perfect-absorption rate? At this value of N , what fraction of the cell's surface is covered by receptors? You may use $R \sim 10 \mu\text{m}$ and $a \sim 10 \text{ nm}$.

This method of approximation, by assuming an intermediate distance between two scales, is an example of the Applied Mathematics technique called *matched asymptotics*.